## Moment-reproduction metric

`moment_reproduction.py`, `compute_moment_reproduction` (2026-11-02)

**Setting.** Each family (a potential: $L_6$, `Scalar_*`, Scat$k$, ...) gives per-signal statistics $\phi_k(x)$. *Imposed* families are the run's own, with its saved fit; *not-imposed* ones (`EXTRA`) are fitted once on the data. Data: the run's training set $\{x_i\}_{i=1}^{n}$; model: the final samples $\{\hat{x}_j\}_{j=1}^{n_s}$ of the run's smallest seed. For each $k$, let $m_k, s_k^2$ (resp. $\hat{m}_k, \hat{s}_k^2$) be the sample mean and unbiased variance of $\phi_k$ on the data (resp. on the samples). Statistics with $s_k^2 \leq 10^{-12}\times$ the family's median $s^2$ are excluded.

**Offset and noise** (in units of $s_k$):

$$\delta_k = \frac{|\hat{m}_k - m_k|}{s_k}, \qquad \nu_k = \frac{1}{s_k}\sqrt{\frac{s_k^2}{n} + \frac{\hat{s}_k^2}{n_s}}.$$

$s_k\nu_k$ is the (unpooled, Welch) standard error of $\hat{m}_k - m_k$ for two independent samples, so $\delta_k/\nu_k$ is the two-sample $z$-statistic; if data and model have the same mean of $\phi_k$, it is approximately $|\mathcal{N}(0,1)|$ for large $n, n_s$ (central limit theorem).

**Verdict** ($K = 2$, $\rho = 0.05$):

| | | | |
|---|---|---|---|
| miss | $\delta_k > K\nu_k$, | | |
| matched | $\delta_k \leq K\nu_k$ | and | $K\nu_k < \rho$, |
| can't tell | $\delta_k \leq K\nu_k$ | and | $K\nu_k \geq \rho$. |

The first condition is a two-sided test at level $\approx 5\%$; the second requires that the test could have detected an offset of $\rho$ data standard deviations. Misses are further classed by $\delta_k$: tiny $< 0.05 \leq$ small $< 0.2 \leq$ clear $< 0.5 \leq$ big (convention). The summary gives, per (model, family), the fraction of statistics in each class.

**Relative error** (table only): $(\hat{m}_k - m_k)/|m_k|$, shown only where $|m_k| \geq 0.5\, s_k$.

**Caveats.** (i) In-sample: an imposed statistic tests the sampler (the SDE corrector), only not-imposed families test the model. (ii) Independence is assumed: the 3 windows of a trajectory are correlated, so $\nu_k$ is somewhat too small. (iii) One seed: seed-to-seed variation of the fitted model is not in $\nu_k$. (iv) Statistics are tested one by one: for a perfect model about 5% of them are misses by chance. (v) For heavy-tailed $\phi_k$ the normal approximation of the $z$-statistic is slow to set in.